

# Vector Electronics

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The purpose of this article is to describe an attempt to solve electronic circuits with non-linear devices like diodes, bipolar transistors, MOSFETs, etc.

A common approach is to set the circuit at some operating point and wiggle around it with the input signal.

For small variations, even the most non-linear device can be approximated by a linear model. The *Small Signal Analysis*, as it is called, is a classical framework for that.

However, some cases are trickier and require the device to operate way beyond the small signal hypothesis assumption. The class AB amplifier is one example: each transistor in the push/pull pair goes from cutoff to linear region as the input changes.

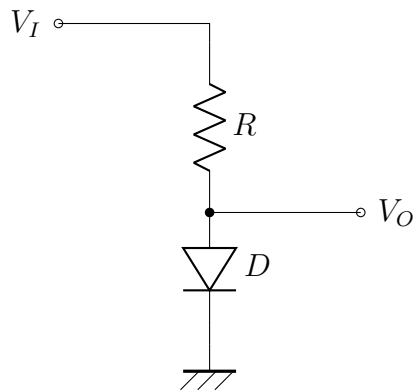
One could try to use a detailed model for the transistor (exponential curve), however:

- the equations usually have no closed-form expression (equations combining  $\exp()$  and polynomials)
- it doesn't give much insight on *how* the thing works.

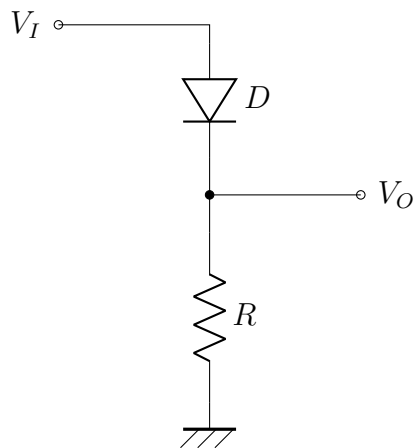
## **An extension of the Small Signal Analysis**

In the following, we will consider that the non-linear device has a characteristic curve that can be modelled as a *piecewise linear function*. From that point on, we will investigate where this leads us.

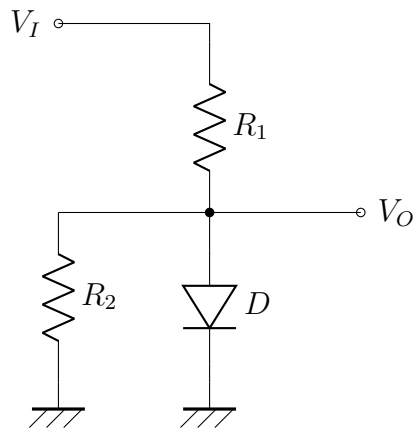
### Example 1: diode and resistor I



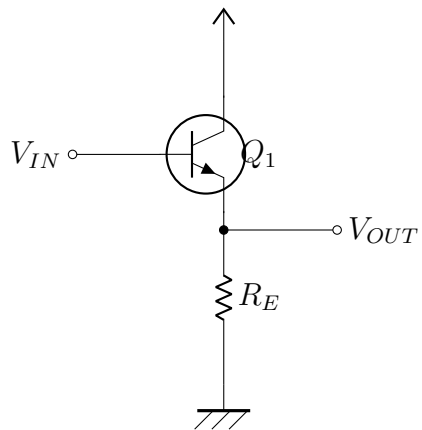
### Example 2: diode and resistor II



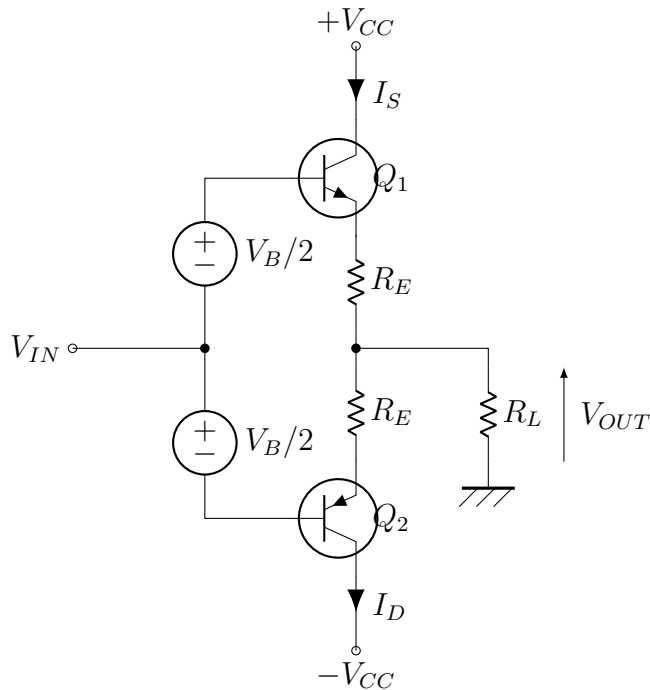
### Example 3: 1 diode and 2 resistors



### Example 4: common emitter



## Example 5: class AB amplifier



**Derivation** Assuming that  $I_C = f_S(V_{BE})$  for the NPN and  $I_C = f_D(V_{EB})$  for the PNP we get the following:

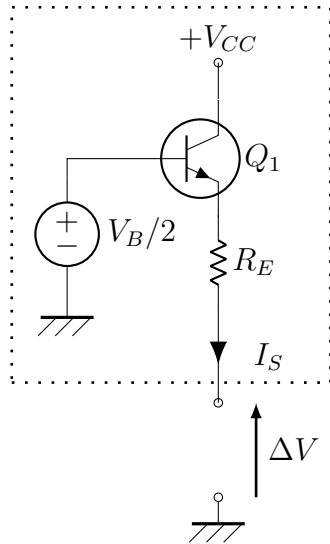
$$\begin{cases} I_S = f_S \left( v_{IN} - v_{OUT} + \frac{v_B}{2} - R_E I_S \right) \\ I_D = f_D \left( -(v_{OUT} - v_{IN}) + \frac{v_B}{2} - R_E I_D \right) \\ V_{OUT} = R_L (I_S - I_D) \end{cases}$$

This time, the equations are trickier: the key variables  $I_S$ ,  $I_D$  and  $V_{OUT}$  are all intertwined. We cannot carry out the analysis the way we did in the previous sections.

Let's try to analyse the system as 2 independant subsystems.

## Class AB subsystem: the current source

If we split the full schematic in half and set the input voltage to a constant, we can simplify the current source of the circuit to a much simpler block:



Using the usual voltage and currents laws gives the following source current:

$$I_S = f_S \left( -\Delta V + \frac{v_B}{2} - R_E I_S \right)$$

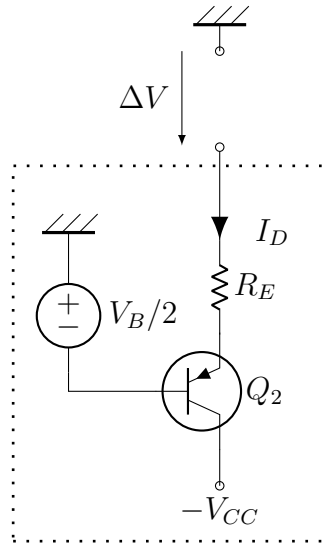
Under a linear model assumption  $f_S(V) = aV + b$ , we can write that:

$$I_S = a \left( -\Delta V + \frac{v_B}{2} - R_E I_S \right) + b$$

Finally, after a bit of algebra:

$$I_S = -\frac{1}{R_E} \frac{a R_E}{1 + a R_E} \Delta V + \frac{a \frac{V_B}{2} + b}{1 + a R_E}$$

## Class AB subsystem: the current drain



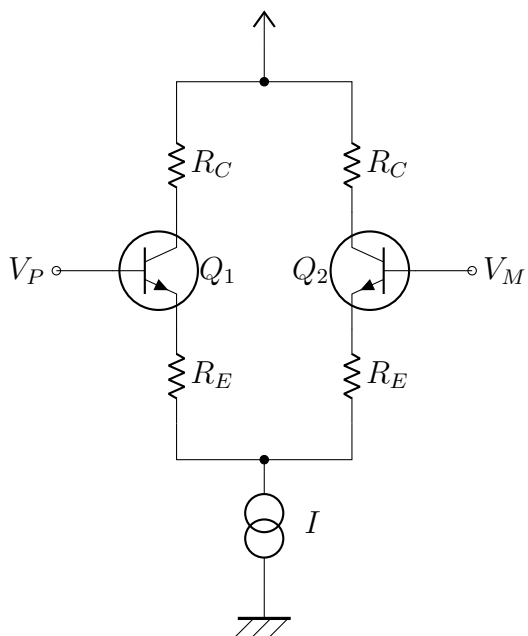
For the current drain subsystem:

Similarly, we get:

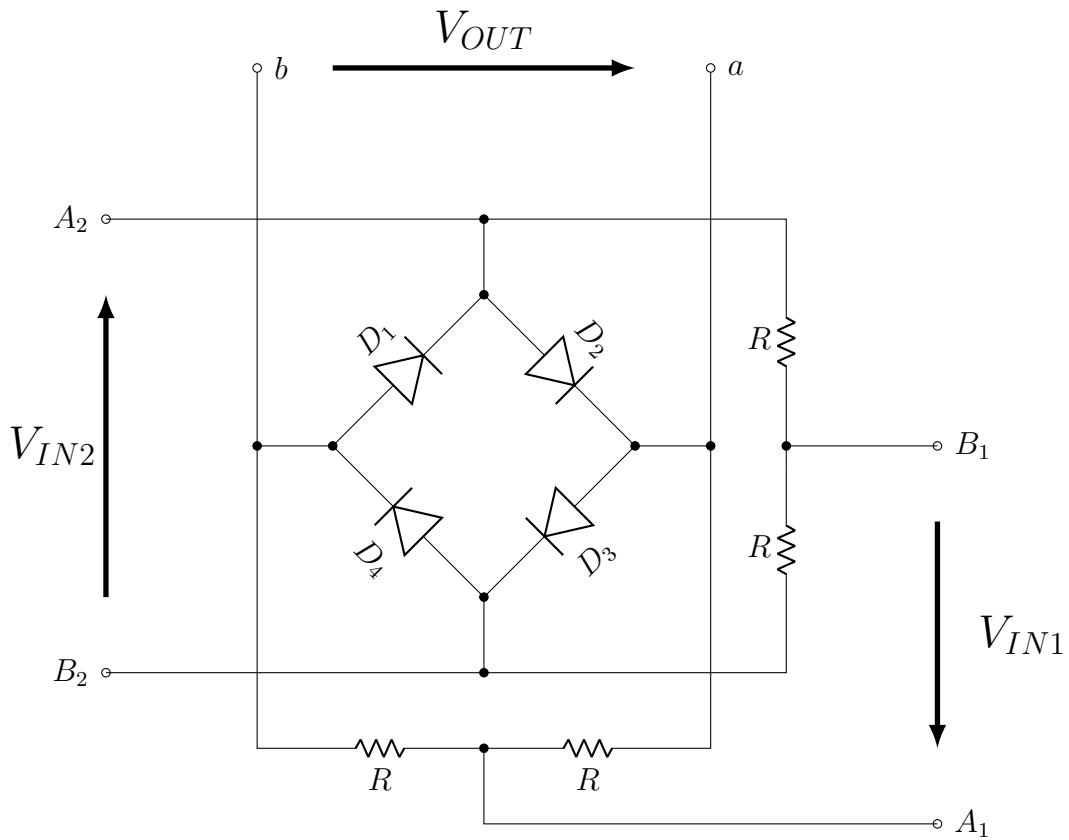
$$I_D = f_D \left( \Delta V + \frac{v_B}{2} - R_E I_D \right)$$

And: 
$$I_D = \frac{1}{R_E} \frac{a R_E}{1 + a R_E} \Delta V + \frac{a \frac{V_B}{2} + b}{1 + a R_E}$$

### Example 6: differential pair



### Example 7: ring modulator



### Example 8: the Gilbert Cell (analog multiplier)

That's a tough one, be patient...